Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Java, C++, Python, SQL, JavaScript

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Multithreading & Concurrency, System Design, Distributed Systems, Design Patterns

Backend & Software Development: RESTful APIs, FastAPI, Flask, Node.js, Microservices, API Design, Authentication (JWT/OAuth), API Testing, Debugging

Frontend: React.js, State Management, HTML, CSS

Databases & Storage: MySQL, SQLite, Redis, SQL Query Optimization

Testing & Software Engineering: Unit Testing, PyTest, Git, GitHub, Docker, GitHub Actions, CI/CD, Linux, Agile Development

EXPERIENCE

VOIS – Vodafone Idea Foundation **|LINK** — ***Aug – Sep, 2025***

- Engineered a modular data processing pipeline to preprocess, analyze, and query two real-world datasets containing 8,500+ records using LLM-powered natural language interfaces.
- Designed and implemented automated validation and response extraction workflows, improving the consistency and reliability of generated outputs across diverse user queries.
- Optimized data processing by integrating preprocessing, prompt generation, response parsing, and result validation into a reusable workflow, reducing manual analysis effort and improving scalability.

PROJECTS

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

API Rate Limiter Library **|LINK**

- Built a production-ready Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for reliable API traffic control.
- Designed a modular architecture with pluggable storage backends and thread-safe concurrent request handling, enabling flexible deployment across application environments.
- Developed 30+ PyTest tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for public installation and reuse.

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Built a full-stack expense management platform using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP-based parsing to automate receipt processing and improve processing time by 40%.
- Engineered backend APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export, enabling secure and reliable application workflows.
- Deployed the application across Render and Vercel, managing frontend-backend integration, authentication flows, database persistence, and production configuration for reliable end-to-end operation.

Customer Lifetime Value Engine **|LINK**

- Developed a data processing and forecasting pipeline over 1M+ retail transactions, using BG/NBD and Gamma-Gamma models to predict 12-month customer value and churn risk for 4,000+ customers.
- Built and deployed Power BI dashboards and a Streamlit-based ROI simulator, transforming model outputs into reproducible analytics workflows for segmentation, risk monitoring, and business decision-making.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** **2026**
- AWS APAC Solutions Architecture virtual experience program | **LINK** **2025**
- Accenture Software Engineering Job Simulation | **LINK** **2025**